fields' force lines, and how they influence our brane, and how fields in our brane influence them. (For more explanation see *Some Technical Notes* at the end of this book.)

When the Professor and his team speak of "solving his equation," in my extrapolation they mean two things. First, figure out the *right* forms for all these things they are guessing: "$U(Q)$, $H_{ij}(Q^2)$, W_{ij}, and $\mathcal{M}$(standard model fields)." Second (following the well-known procedure), deduce, from his equation, everything he wants to know about our universe, about the anomalies, and most important, about how to control the anomalies so as to lift colonies off the Earth.

When characters in the movie speak of "solving gravity," they mean the same thing.

In the movie, when the Professor is very old, we see him and grown-up Murph trying to solve his equation by iterations. On a blackboard, they make a list of guesses for the unknown things (guesses that I wrote on the board just before the scene was filmed; Figures 25.8 and 25.9). Then, in my extrapolation, Murph inserts each guess into a huge computer program that they've written. The program computes the physical laws for that guess, and those laws' predictions for how the gravitational anomalies behave.